

A wide-angle photograph of the Miami skyline across a body of water under a dramatic, cloudy sky. The skyline is composed of numerous high-rise buildings of varying heights and architectural styles, including modern glass skyscrapers and older concrete structures. The water in the foreground is a deep blue-green with gentle ripples. The sky is a vibrant blue, filled with large, white, puffy clouds that catch the light, creating a high-contrast scene. The overall mood is one of a bustling, coastal metropolis.

Week 1 City of Risk/City of Retreat: the case of Miami Florida



- **Week1 City of Risk/City of Retreat: The case of Miami, Florida**
- **Why is Miami rated the 1st in the world in terms of assets vulnerable to sea level rise?**
- **Can Miami future proof itself against rising water? Can it win over public support and retro-fit its buildings and infrastructure against the inevitable threats of climate change?**
- **Should we say 'Goodbye, Miami' because it's a Doomed City? The call to RETREAT!**
- **Is it a disposable city as some ecologists argue? And what does disposability entail?**

Climate change may be a defining issue of our time—at least that is what the UN declares but **are we really living in an era of Anthropocene.?** What does that mean? And when did it start? Have we reached a ‘tipping point’?

Is the Anthropocene era defined by **an era of superstorms and wildfires, rising sea levels** and depleted water sources, heat waves, crop failures, polar vortexes, bomb cyclones, climate migrations and decreased biodiversity?

Are we really in a life-or-death struggle for the Earth’s and therefore life’s survival?

Can we do anything about these global trends?

With a great deal of **uncertainty surrounding the rhetoric we need to be certain what these terms mean, what they render invisible**, why they intend to **shock** not to aid the development of alternatives.

I am concerned about **the rhetoric surrounding the discussion of cities in the 21st Century**. And so, this course is organized around **the rhetoric of urban buzz words – which have become like names on a map of discovery. But what do they mean?**

We start out with “**Anthropocene**’ and the new buzzword ‘**Retreat**’ in the case study **Miami --- the number 1 city threatened by rising seas in the 21st century**.



THE CASE OF MIAMI BEACH, MIAMI and MIAMI-DADE COUNTY, Florida

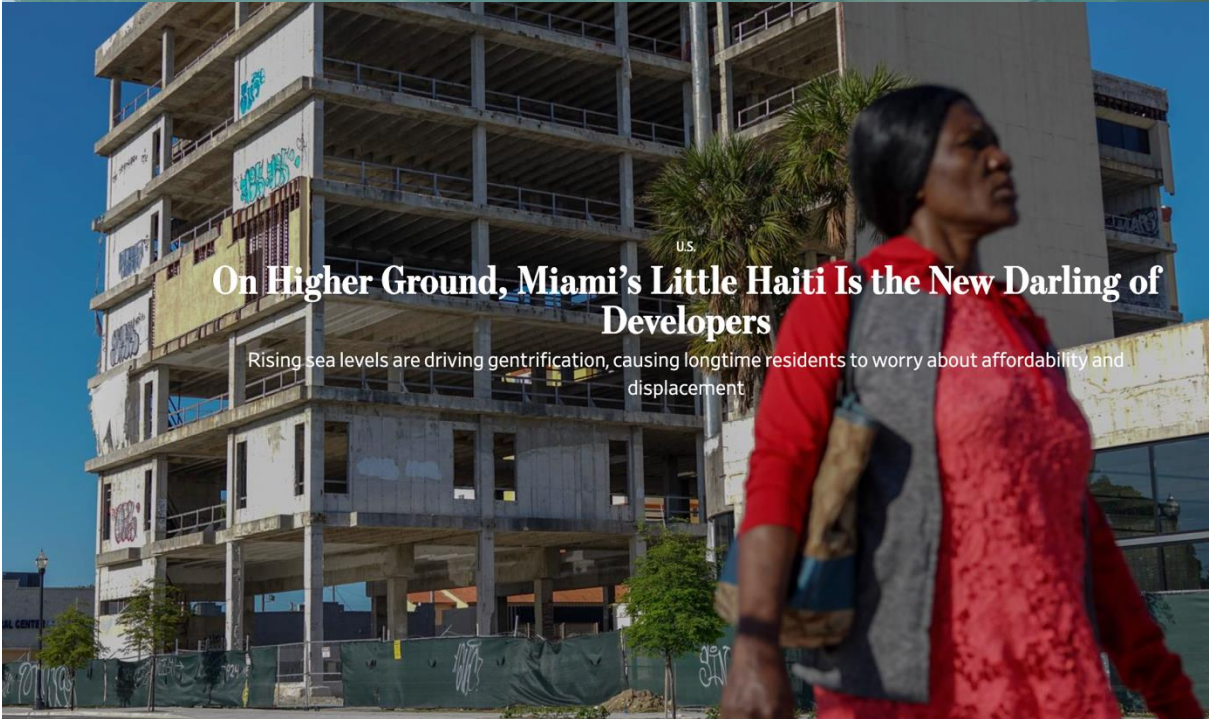
- I have long been fascinated by Miami **Florida because it is a city built on porous limestone** –a subterranean plateau with holes in it likened to **Swiss cheese**.
- This substructural base lets **seawater and freshwater seep up from below ground** and flood the city, rather than limit its inundations to downpours or storm waves that clog the drains and cause the streets to flood. **This sponge-like substructure means water can roam about freely, seeping into yards at high tide, bubbling up onto streets even when it is not raining, and corroding foundations from below.**
- Miami is as well one of **the lowest lying Metropolitan districts in the USA**. In a world of rising seas, Miami is in big trouble and has become a **posterchild city for tackling this rise**.
- In fact, **Miami flooding** is projected, so says Stephanie Wakefield a resiliency scholar, to occur **380 times per year by 2045**.



- Whatever your imagination of Miami/Miami Beach may be -- **a magical city, a sunny paradise, an apocalyptic city, a doomed city** – it is a city that is ranked **first in the world in terms of assets vulnerable to sea level rise**. Its wealth is dependent on **real estate taxes and tourist spending** and thus its managers and developers exist in constant fear that **this income stream will be threatened by climate change**.

- Hence Miami has become **a laboratory** for climate adaptation. It has promoted a program of **resiliency infrastructure** called “**Miami Beach Rising Above**” which consists not only of **elevating low-lying streets** but also deploying a series of **new pumps** to push the water out and **seawalls** to keep the water at bay. There are, in addition, a variety of resiliency programs put in place across metropolitan Miami not just in Miami Beach.

- The region is called a ‘**crisis city**’ perpetually **vulnerable** to environmental risks, constantly **surveilled** and **managed** with a bricolage of resilient infrastructures protecting it against all and every ecological disaster. **Resiliency planners believe the city can be saved!**



On Higher Ground, Miami's Little Haiti Is the New Darling of Developers

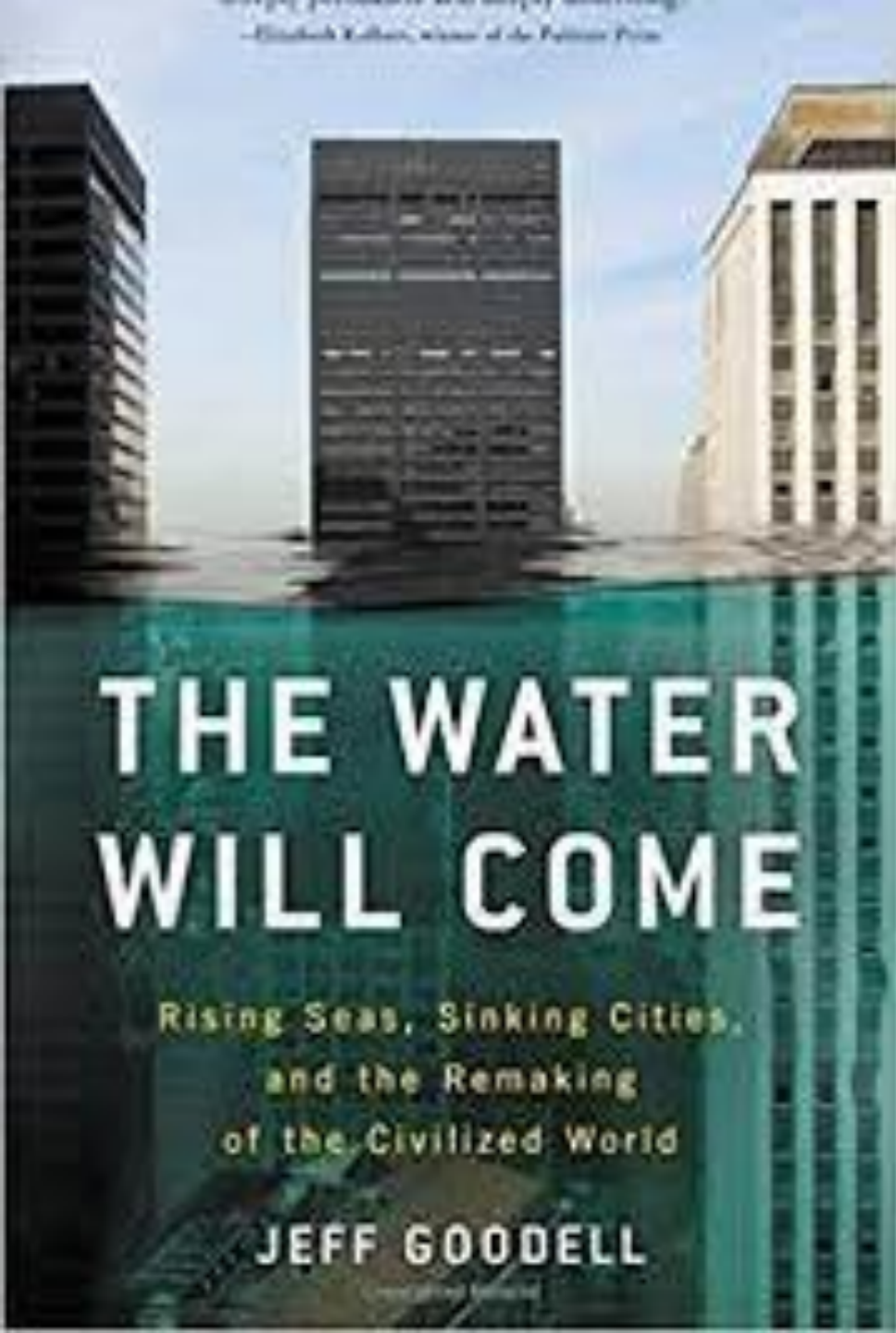
Rising sea levels are driving gentrification, causing longtime residents to worry about affordability and displacement

- Since low-lying Miami is the most exposed American metropolitan area to sea-level rise, the programs implemented in Miami will become **case studies for other cities and regions facing climate challenges**.
- Because the city has the most expensive coastal real estate in the world, it has an **ample tax base to experiment** with different solutions, and it is expected there will be many.
- It may **elevate houses** on stilts so that flood waters can pass underneath; it could truck in tons of dirt and rocks from elsewhere to **raise the level of the ground**; and it could build **denser housing** on higher land away from the ocean and it could retreat from the lowest land to higher land.
- These are **all costly procedures**, and of course there is the question of who is going to pay from them? Most are aware, however, that **pushing out the water** by building an elaborate network of sea walls won't work to solve the problem of **water seeping up through its limestone substructure**.
- Moving to higher ground has community costs and leaves the most vulnerable such as the elderly behind.

Others are less optimistic about Miami's resilient future. The journalist Jeff Goodell, author of “Goodbye, Miami” (2013) and former Vice President Al Gore featured in the documentary film *an inconvenient sequel Truth to Power* (2016) both see **Miami's shrinking future confronting sea level rise as an apocalypse from which the city will not survive.** There is no way to stop water rising up through its porous geology; surrounded by water on all sides, the city is like a sponge in water. It must retreat!

Goodell wrote “Goodbye, Miami” a 2013 article in which he begins with a scene of the city succumbing to a fictional **Hurricane Milo** in 2030. He writes:

“When the water receded after ..., there was a foot of sand covering the famous bow-tie floor in the lobby of the Fontainebleau hotel in Miami Beach. A dead manatee floated in the pool where Elvis had once swum. Most of the damage occurred not from the hurricane's 175-mph winds, but from the 24-foot storm surge that overwhelmed the low-lying city. In South Beach, the old art-deco buildings were swept off their foundations. Mansions on Star Island were flooded up to their cut-glass doorknobs. A 17-mile stretch of Highway A1A that ran along the famous beaches up to Fort Lauderdale disappeared into the Atlantic. The storm knocked out the wastewater-treatment plant on Virginia Key, forcing the city to dump hundreds of millions of gallons of raw sewage into Biscayne Bay. Tampons and condoms littered the beaches, and the stench of human excrement stoked fears of cholera. More than 800 people died, many of them swept away by the surging waters that submerged much of Miami Beach and Fort Lauderdale; 13 people were killed in traffic accidents as they scrambled to escape the city after the news spread – falsely, it turned out– that one of the nuclear reactors at Turkey Point, an aging power plant 24 miles south of Miami, had been destroyed by the surge and sent a radioactive cloud over the city.”



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- Goodell's disaster scenario depicting many of the risks that Miami currently faces, **did not amuse local officials or managers** intent on building a resilient Miami against the future of rising seas.
 - Nevertheless, Goodell's story **is a warning and a call for action before it's too late**. Goodell's story is also a diatribe against **modern humankind's hubristic – and Anthropocentric -- belief that technology can triumph over nature**;
 - that a mosquito-and-crocodile infested **swampland could be tamed**; that canals and pumps would drain the swamps; that **human ingenuity in the future will triumph** over any threatening hurricane no matter how violent and destructive its winds, waves and storm surges may be.
 - Miami is **a manufactured landscape** based on exploitation of its natural environment and it cannot survive if this **nature/human exploitation** continues. **The city has to think otherwise. [this is a normal apocalyptic evaluation of the environmental crisis cities face but as stated above, city managers disagree.]**

- The jury, however, is still out **pondering whether Miami will survive** to the end of the 21st century. If geologists' projections are correct, the sea by then could reach a **ten-to-thirty-foot rise**.
- The **average** elevation of Miami-Dade County is just **six feet above sea level** and **half** of the area surrounding Miami is **less than five-feet** above sea level, thus a six-foot rise would cover almost everything.
- Clearly the city is threatened with **drowning**. Can it **future-proof** its storm-water system and its public and private infrastructure against climate change?
- Can it win over **public support** for all the retrofitting of buildings, the elevation of roads, the **rewriting of building codes and land use laws that will be necessary?**

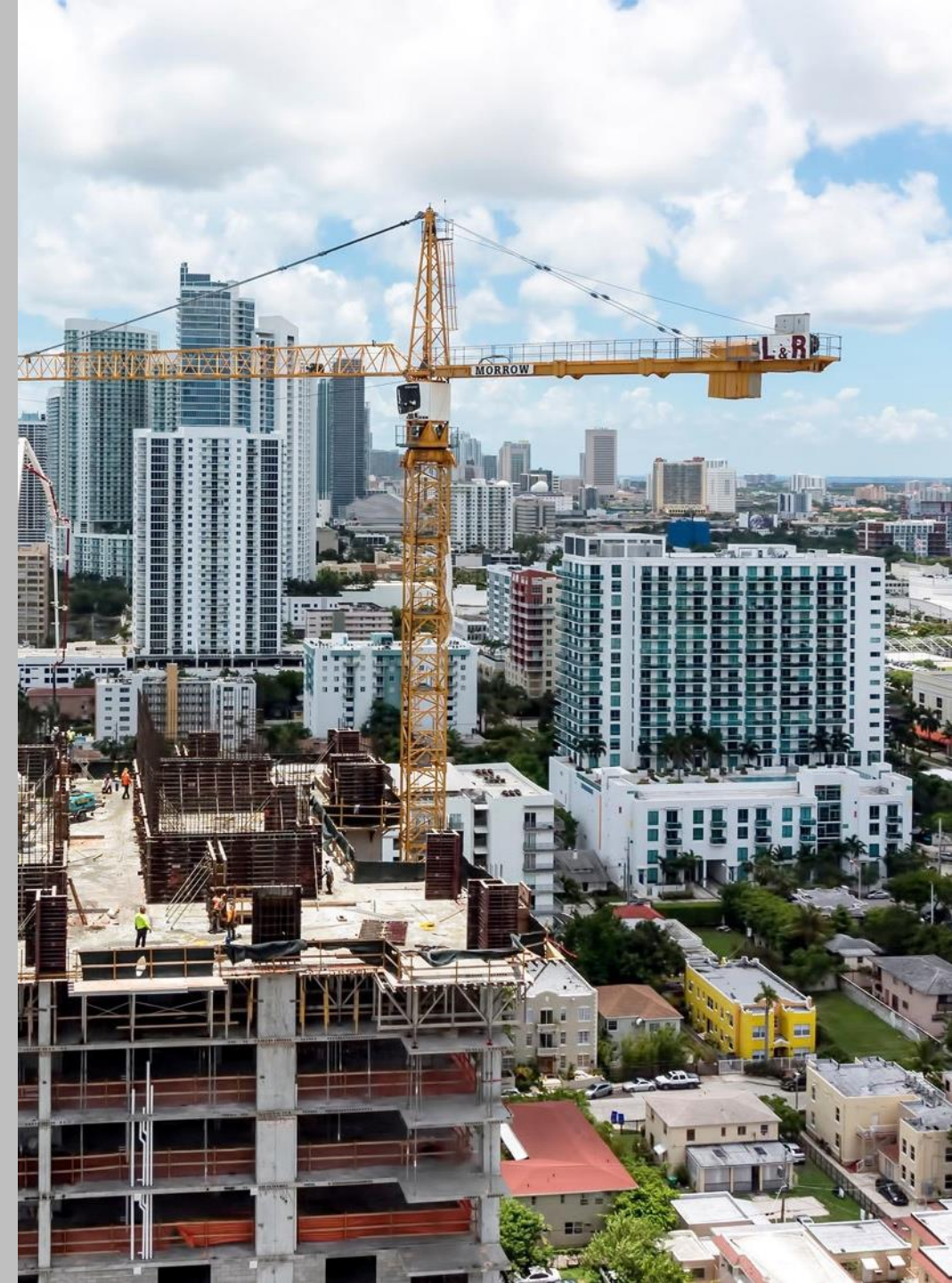




- These **doom sayers**, and there are many others, have been warned not to say anything that will **sully the city's reputation as a vacation paradise**. Tourism depends on clean water to swim and bath in; tourists won't come to visit the city if they think the water is contaminated.
- Goodell is provocative: "**Leaky septic systems** are not the only source of contamination urban residents need to be concerned about when their city starts to flood.
- In Miami, a 200-acre garbage dump known affectionately as **Mount Trashmore sits right on the edge of Biscayne Bay**""The disposal pit is sealed off by layers of clay at the bottom, but this stew of arsenic, chromium, copper, nickel, iron, lead, mercury, zinc, and benzene was **never designed to be submerged.**"

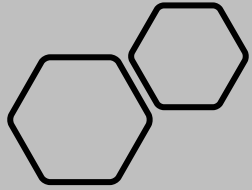
LAND MATTERS:

- **Real estate developers** reign supreme in Miami-Dade County and to repeat, its most valuable real estate is located **right on the water**.
- The Organization for Economic Co-operation and Development [OECD] lists Miami as the **number-one most vulnerable city worldwide in terms of property damage**, with more than \$416 billion in assets at risk to storm-related flooding and sea-level rise.
- For the most part, its **developers deny there is a climate crisis**, or they believe that the resiliency plans for the city are sufficient protection.
- Developers **cynically go about developing more high-rise condominiums** selling them off before construction is completed so they really don't need to be concerned about the **gravity of the city's future**.
- Are Climate threatened cities caught in a cycle of Disaster-Rebuild-Disaster-Rebuild or can they design policies to Retreat?

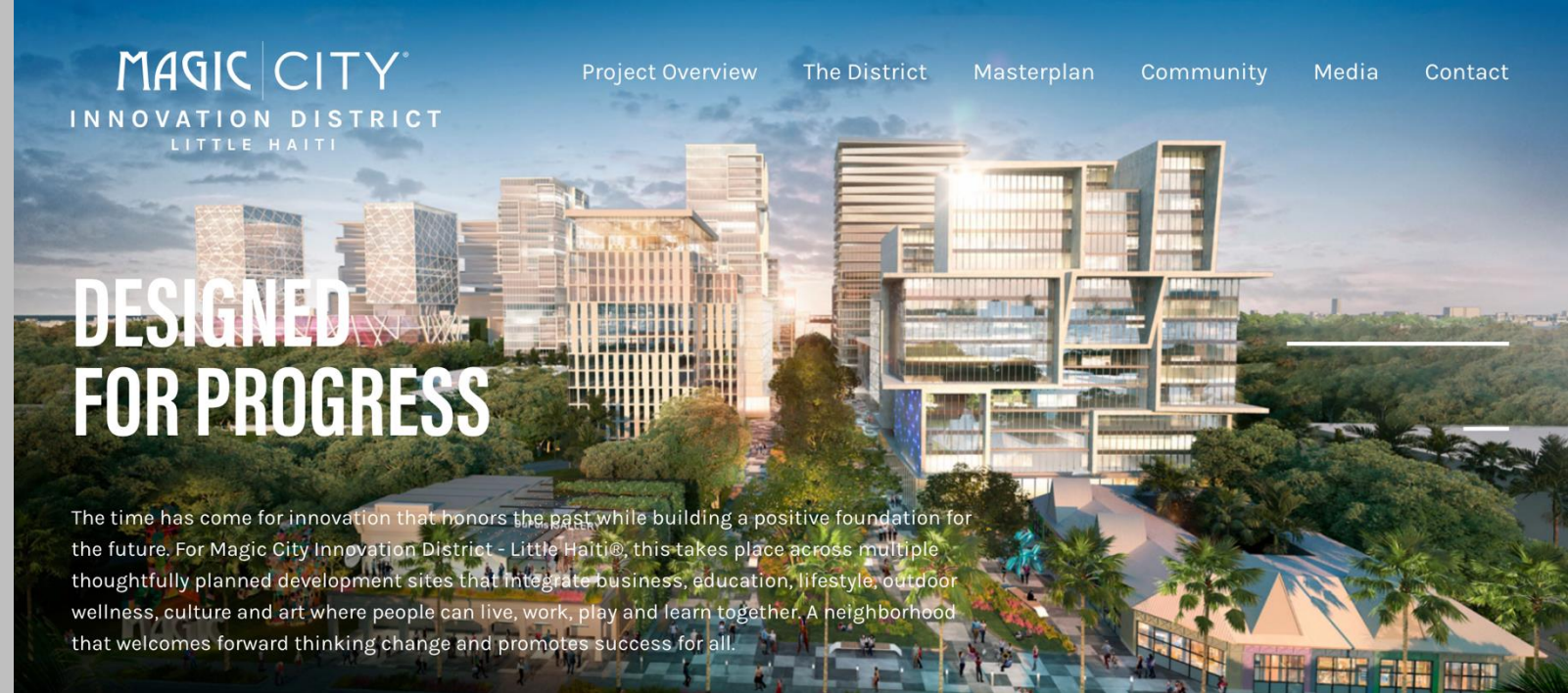


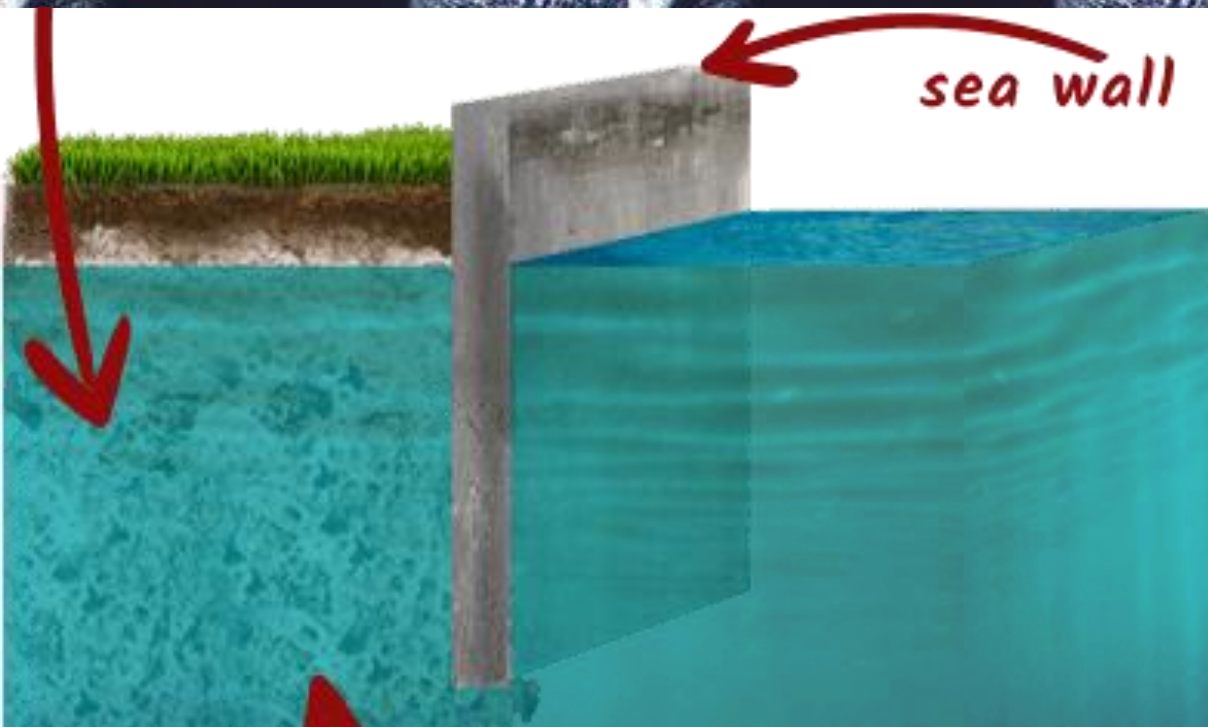


- While Miami may be a very rich city, it is a **manufactured landscape divided between rich and poor**. “**Forty percent** of the households in Miami-Dade County are working poor, people with little savings and few assets. **Nineteen percent**—nearly one-fifth—live below the poverty line.”
- It is a landscape that has been shaped by a history of **red-lining, racial segregation, urban renewal, and highway construction**.
- Most working-class and immigrant communities are located along the inland **Atlantic Coastal Ridge** whose **elevation** is twice that of the surrounding swampland that has been drained long ago and where the rich and better-off tend to dwell.
- For example, the Miami neighborhoods of **Little Haiti, Liberty City, Overtown, Allapattah and Little Havana**, are located not only inland from the high-cost areas along the coastline but also on some of the highest ground and have historically been home to **minority immigrant groups**.



- If the city is to survive, so Mario Ariza argues in *The Disposable City*, it will have to **cluster high-density mixed-use development on this high ground.**
- In a real-estate driven economy, this inevitably will generate **climate gentrification** as the mostly black and Latinx minorities who live in these higher elevation communities will be pushed out by higher-priced development onto lower ground more prone to flooding.
- Take **Little Haiti** for example. Here higher priced housing is replacing that of lower priced housing, driven – it is assumed -- by the **nemesis of sea-level rise**. Real-estate speculators know exactly what they are doing by playing into consumer preference for **less risky investments**. The massive and controversial **\$1billion Magic City Innovation District** has begun its phased construction in Little Haiti.





- Accepting Miami as a resilient city or an apocalyptic disposable city, there is yet **a third vision of Miami's future.**
- Many Miami planners and politicians accept that the city's **resilient infrastructures** are insufficient; not exactly a waste of money but **they will not eradicate the city's vulnerability.**
- Despite all the efforts of mitigation, the city will still **flood from below** because of its porous foundations. It is no longer a matter of '**if**' the city will survive a killer storm, but a matter of '**when**' it will come. **It is time to think innovatively even utopianly.**



- “**South Florida Islands**” so named by **Tom Gustafson** envisions a future in which Miami will be dramatically restructured into a **series of island zones** raised on high ground, connected through **elevated bridges**.
 - Gustafson, a former speaker of the Florida House of Representatives believes Miami is doomed even with a six-foot sea rise, and that most of South Florida is ‘toast’.
- “You can’t even say the words ‘climate change’ on the House floor without being run out of the building” Gustafson moaned a decade ago. The future of Miami means the population will need somewhere to migrate to.
- He advocates demolishing the present-day city, using the **debris as fill for** totally new elevated islands and bridges and all structures built with extreme **storm-resistant materials**. Most of the city’s former population will be **evacuated** to other regions: only the **denizens of luxury towers and service providers will remain**. The islands will become **self-sufficient in food, materials, manufactured goods, and energy**.



- This proposal is no simple task: it will require many different systems of **transportation, freshwater collection and delivery, solid waste and wastewater disposal, solar and wind energy generation, new educational systems, and job creation programs.**
- It calls for new methods of **growing food locally, of procuring materials for local production and construction, of planting useful trees on the islands and developing plantations filled with mangrove forests in spaces between the islands.**
- Everything a city and its population need to survive in this **new climate of severe storm systems**, will have to be meticulously thought-out ahead of time and their provision delivered equitably and justly. **It will require land-use PLANNING**



- **Accepting that Miami will drown in the not-so-distant future, calls forth other futuristic proposals to save the city.**

- **John Stuart**, the chair of the architecture department at Florida International University, is working with students and professors on a **multiyear project** to imagine what South Florida's future might look like.

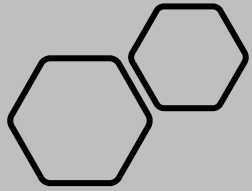
- The driving question is if you can't stop water from entering the city, **how do you learn to live with it?** They draw inspiration from **Stiltsville**, an **abandoned collection of structures built-on pilings in Biscayne Bay** in the 1930s of which a few structures still survive.

- **Can this be replicated in the 21st century but how do you build a floating city with all the services a city requires?**

- Stuart ponders on the future of Miami: "... if sea-level rise happens **slowly enough and Miami doesn't get hit with a hurricane and the drinking-water supply doesn't go bad and the real-estate market doesn't crash and the beaches aren't washed away, the city of Miami may well have time to transform itself into a modern Venice.**"



- A more pragmatic future of working with nature is presented by Walter Meyer and Jennifer Bolstad of **Local Office Landscape Architecture (LOLA)**. They are focused less on long-term solutions and more **on buying time** and encouraging “city planners to think about the **natural landscape as something more than just a pile of dirt to be bulldozed and paved over.**”
- Working in the north Miami community known as **Arch Creek Basin**, they propose tearing down houses that are currently built on paved over **sloughs (meaning natural channels** or ‘transverse glades’) in the low-lands that once connected the Everglades to the sea.
- Because the area frequently floods, **insurance companies** mark these properties as ‘**repetitive loss properties**’ and deny them access to flood insurance coverage.



- Instead, Meyer and Bolstad would **build new housing on higher ground** and restore the system of natural drainage allowing water to collect and drain along the contours of the land.

- Calling their landscape efforts '**forensic ecology**' because the Latin word 'forensis' means '**working in public**' and they intend their environmental recovery solutions to include **community land banks, collective housing** and more. They are working with the community, keeping it in place.

Future Slough: Balances people, water, nature



Innovative Solutions for Miami doomed city:

Arch Out Loud, an architecture research initiative which manages competitions mounted a call for “**Miami Floating Housing**” in 2022. The call was for innovative designs addressing both Miami’s housing crisis and the threat of sea level rise. The winner was Gabriele Filippi Architetto’s “**Green Water Village**”. The design imagines a network of biophilic islands located in Biscayne Bay, just south of Miami. [images to the right]

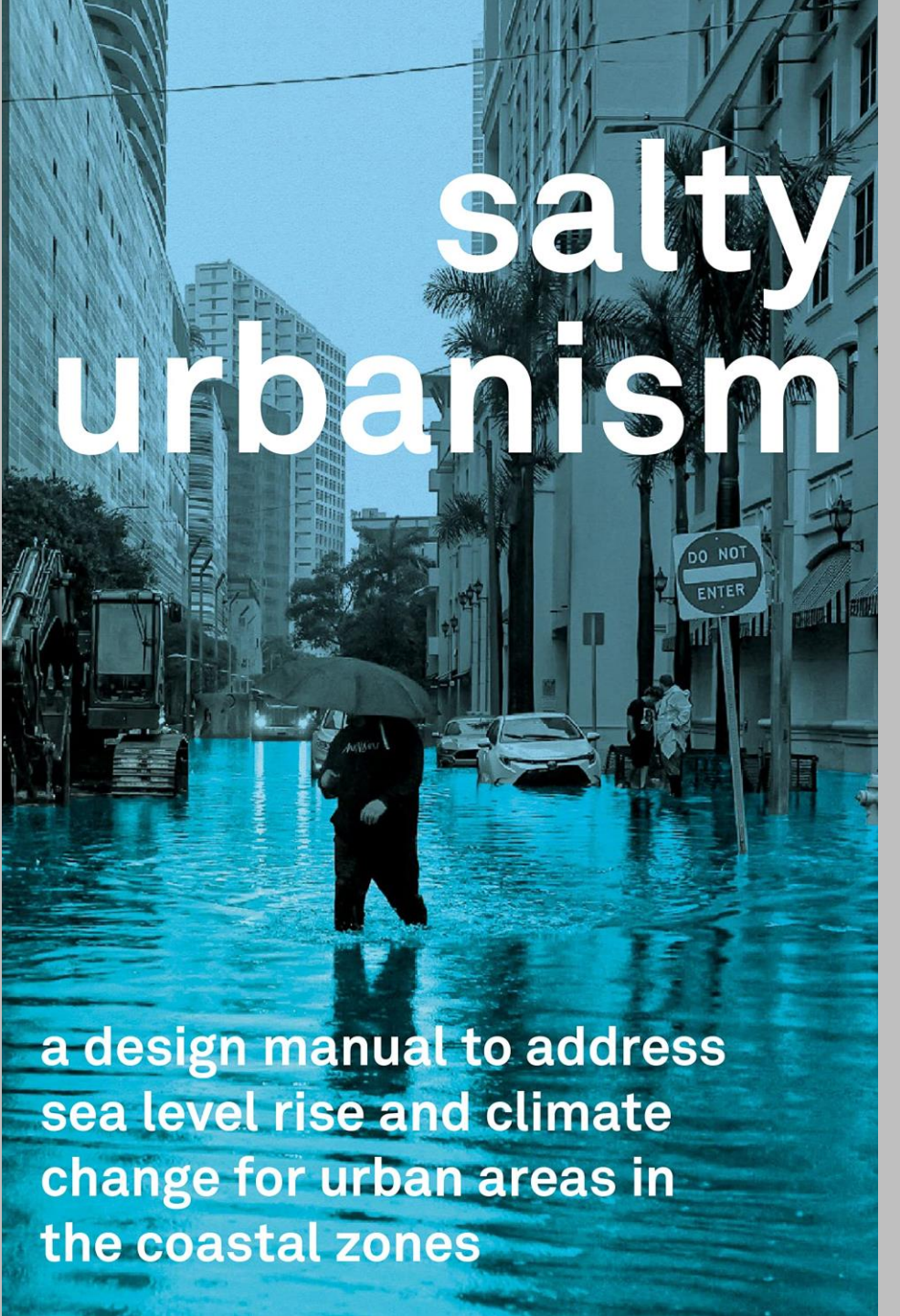
The prefabricated houses constructed of metal, wood, and glass, have irregular shapes that host gardens and communal meeting spaces. An aggregation of **these island houseboats creates small ports**. A continuous and sinuous wooden path connects all the units.



“Salty Urbanism”, a design manual addressing sea level rise along coastal zones in south Florida, written by architect Jeffrey E. Huber in 2024, addressing how we live on and with water. It offers **three design scenarios**: 1/ **Soft Defense**—designing living shorelines and break waters, blue and green streets with salt-tolerant plants, pervious pavements, bioswales and rain gardens. Adapting some of New Orleans schemes for coastal cities. 2/ **Strategic Retreat** – removal of low-lying communities to higher lands on the coastal ridge, like Little Havana in Miami. The abandoned low-lying lands will be **rewilded**. 3/ **Land Adjustment** proposing amphibious architecture, elevated and submerged housing, and island zones. The latter will contain localized production of food, water and waste management and it extends to the planning of water taxis and ferries, and perhaps energy supplies as well. This latter scenario will require **“Unplanning Miami”** no longer thinking of making cities resilient against flooding, but **dismantling** them entirely, using the **pulverized former land for a new geography of land filled islands**. This scenario is similar to what **Gustafson designed**. This means **Retreat** to island territories.

Jeffrey Huber is a member of the LA architecture firm Brooks + Scarpa .

salty urbanism

A photograph of a flooded city street, likely in Miami, showing a person walking through the water holding an umbrella. A "DO NOT ENTER" sign is visible on the right side of the street. The image is used as a background for the title and subtitle.

a design manual to address sea level rise and climate change for urban areas in the coastal zones

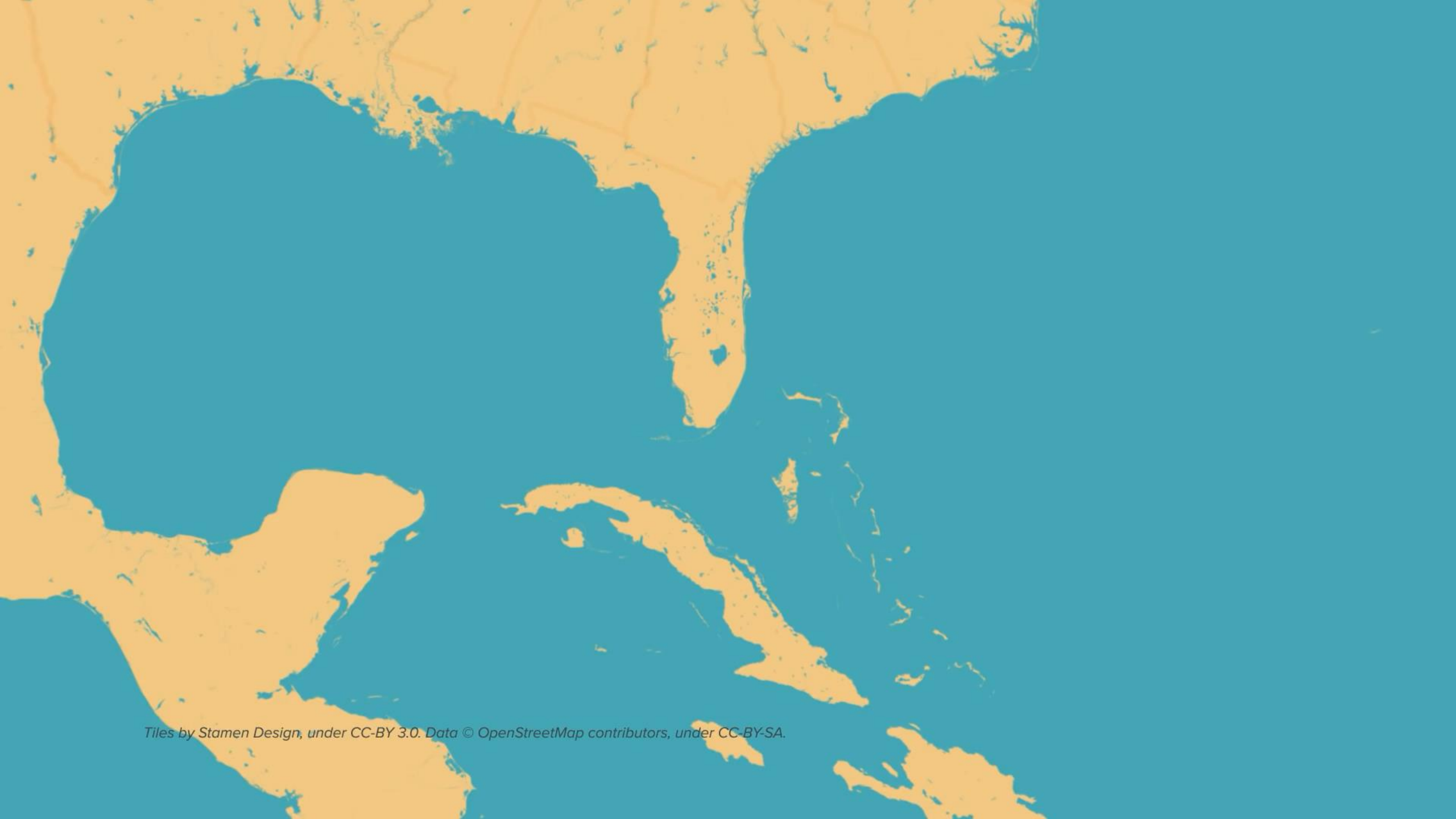
- More likely than any of these scenarios, the **ocean will seep slowly into the city**, reaching higher levels and more frequent invasions every year, until **‘the’ big storm** arrives and devastates the place. **Maybe then**, people will start to question the wisdom of living in Miami and begin **to flee rather than flow toward** the drowning city.

- But it won't be a **peaceful transition!** Once the impact of climate gentrification is fully felt and the city's **simmering history of racial discrimination boils over** as the water rises – there will be a **fight over who goes and who stays.**

- This is an **apocalyptic beginning** for an introduction to Urban Studies but cities in the 21st century are **facing crises, which they must solve, or they may be transformed dramatically, or some say perish.**



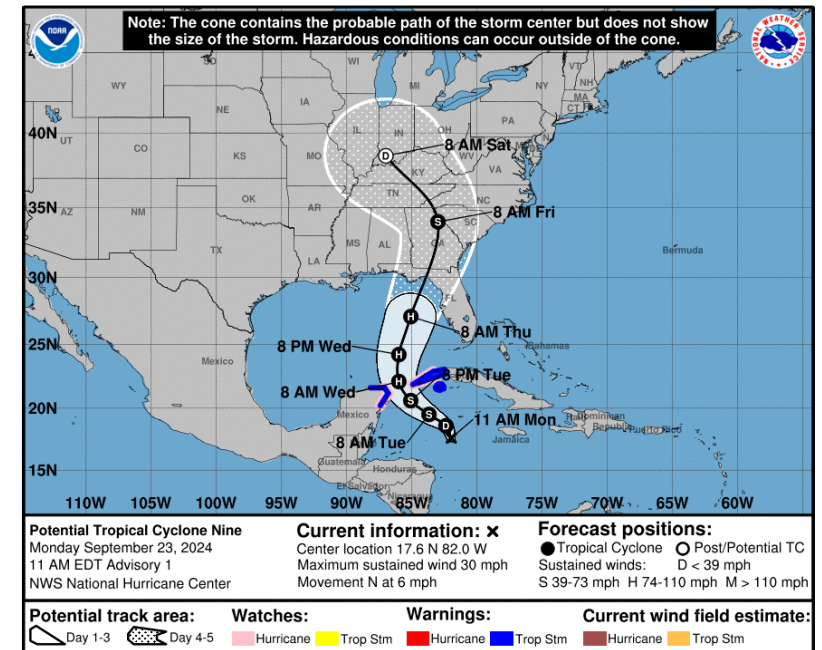




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THE CALL TO RETREAT FROM THE SOUTHERN USA

- Abraham Lustgarten, “The People Fleeing Climate Disasters Are Going to Transform the American South.” *The New York Times* (Oct. 2, 2024).
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- **Hurricane Helene**, barreling through Florida’s coastline and then up into the Carolinas in late September 2024, was a 420-wide slowing moving **storm of wind and water**. It struck down residents’ faith in the rationality of residing in the **Southern USA**.
- How many times would they be **embattled by sea level rise, extreme heat, and soaking rains?** How many times will governments, federal, state and local, **bail their citizens out even as the costs of recovery keeps increasing?** How many times will they face a **disaster-rebuild-disaster cycle?**





- It is now estimated that tens of millions of Americans will eventually **move away from disaster zones in the south** to safer places inland or in northern cities. One study estimates that **3.2 million Americans have already migrated to higher grounds** away from flood zones, even if only short distances, moving for example away from low-lying parts of Staten Island, Miami and Galveston. And perhaps another **7.5 million will migrate over the next 30 years.**

- Those who move may be the **young, middle class and mobile** leaving the poor and elderly behind and exacerbating the wealth divide in their wake. The population left behind are those which require **greater services, medical attention, low-cost accommodation.** These needs will be met with diminished revenues as communities shrink and recovery budgets shrink.

To be continued.....